

■ 3.4.7 Solving Equations with Subsidiary Conditions

In doing calculations with equations, you will often find it convenient to think of a particular set of equations as the “main” ones you are working with, and to think of other equations as “subsidiary conditions” that must also be satisfied.

A typical thing to do is to assign a name to the list of subsidiary conditions, and then to include them by name in each list of equations you give to `Solve`.

`sincos` is defined to be the equation
 $\sin^2(x) + \cos^2(x) = 1$.

```
In[1]:= sincos = Sin[x]^2 + Cos[x]^2 == 1
```

```
Out[1]= Cos[x]^2 + Sin[x]^2 == 1
```

This solves the equation
 $\sin(x) + 2\cos(x) = 1$, with the “subsidiary
 condition” $\sin^2(x) + \cos^2(x) = 1$.

```
In[2]:= Solve[ { Sin[x] + 2 Cos[x] == 1, sincos } ,  
               { Sin[x], Cos[x] } ]
```

```
Out[2]= {{Sin[x] -> -(3/5), Cos[x] -> 4/5},
```

```
         {Sin[x] -> 1, Cos[x] -> 0}}
```

Here is another equation, solved with the
 same “subsidiary condition”.

```
In[3]:= Solve[ { Sin[x] == Cos[x], sincos } ,  
               { Sin[x], Cos[x] } ]
```

```
Out[3]= {{Sin[x] -> 1/Sqrt[2], Cos[x] -> 1/Sqrt[2]},
```

```
         {Sin[x] -> -(1/Sqrt[2]), Cos[x] -> -(1/Sqrt[2])}}
```

Section 3.4.10 discusses another way to set up and use algebraic subsidiary conditions.